

Activity - Scenario Based Problems

1.

Technique used:

Median Filtering

Reasons:

Removes salt-and-pepper noise effectively while preserving important edges.

2.

Technique used:

Gaussian Filtering

Reason:

Smooths Gaussian-noise and reduces sensor-related variations.

3.

Technique used:

Bilateral Filtering

Reason:

Reduces noise while preserving important edges and fine structures.

4.

Technique used:

Median Filtering

Reason:

Removes isolated black and white dots without significantly affecting the text.

5.

Techniques used:

Gaussian Filtering

Reason:

Smooths the grainy noise and produces a cleaner image.

6.

Techniques used:

Histogram Equalization

Reason:

Improves contrast and makes road features more visible.

7.

Techniques used:

CLACHE

Reason:

Enhances local contrast and improves visibility in dual underwater images.

8.

Techniques used:

CLAHE

Reason:

Enhances local details and makes internal structures more distinguishable.

9.

Techniques used:

contrast stretching

Reason:

Expands the intensity range and improves visibility of objects.

10.

Techniques used:

contrast stretching

Reason:

Expands the narrow intensity range to improve overall contrast.